$$g(x,y) = f(x,y) * \underbrace{h_1(x,y) * h_2(x,y)}_{h(x,y)} - f(x,y)$$

h_1

1	1	1

$$G(u,v) = F(u,v)\,[\,H(u,v) - 1\,]$$

h_2

1
0
-1

h

1	1	1
0	0	0
-1	-1	-1

$\} h(x,y)$

$\xrightarrow{F}$

3	0	0
0	0	0
-3	0	0

$\downarrow\downarrow\downarrow$

$H_{12}(u,v)$

0	0	0
	0	0
	0	0

$4,5 - j2,6$ ←

$4,5 + j2,6$ ←

$H(u,v)$

-1	-1	-1
$3,5-j2,6$	-1	-1
$3,5+j2,6$	-1	-1

-1	-1	-1	-1	-1	-1
$3,5-j2,6$	-1	-1	$3,5-j2,6$	-1	-1
$3,5+j2,6$	-1	-1	$3,5+j2,6$	-1	-1
-1	-1	-1	-1	-1	-1
$3,5-j2,6$	-1	-1	$3,5-j2,6$	-1	-1
$3,5+j2,6$	-1	-1	$3,5+j2,6$	-1	-1

→ dikeydeki yineleme yönüdür

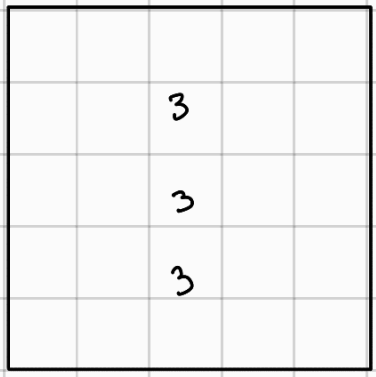
3	1	1
2	7	7
0	2	1

z	pz
0	0
1	2
2	2
3	2
4	1
5	1
6	1
7	0

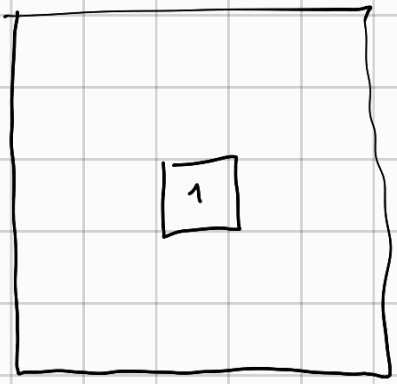
r	$p_r(r)$	$s = T_r(r)$	G(z)	
0	1/9	1/9	0/9	0, 1 → tutarlı olmak için
1	3/9	4/9	2/9	2
2		6/9	4/9	3
3		7/9	6/9	4
4		7/9	7/9	4
5		7/9	8/9	4
6		7/9	8/9	4
7		8/9	9/9	7

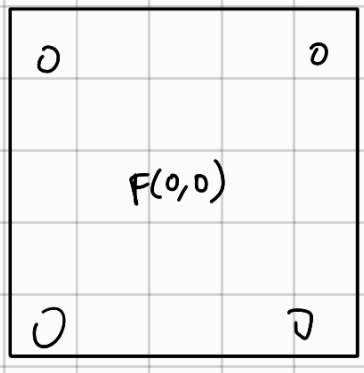
3. soru $(-1)^{x+y}$

F



H





$$f(x,y) = \sum_u \sum_v F(u,v)\, e^{j2\pi\left(\frac{ux}{M}+\frac{vy}{N}\right)}$$

Image Segmentation

10.1 Fundamentals

Let R represent the entire spatial region occupied by an image. We may view image segmentation as a process that partitions R into n subregions, $R_1, R_2, \ldots, R_n$, such that

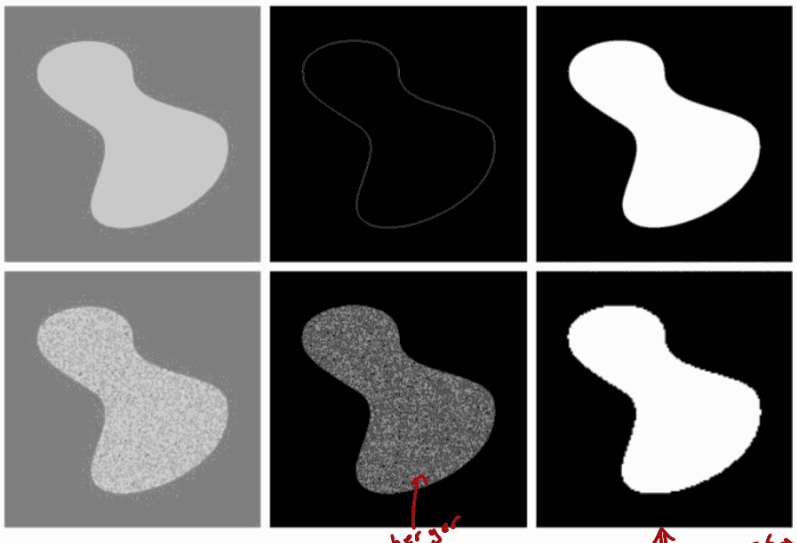
(a) $\bigcup_{i=1}^{n} R_i = R$. → Tüm bölgelerin toplamı image'ı kapsıyor olmalı

(b) R_i is a connected set, $i = 1, 2, \ldots, n$.

(c) $R_i \cap R_j = \varnothing$ for all i and j, $i \neq j$. → Her bölgenin kendi pikselleri var

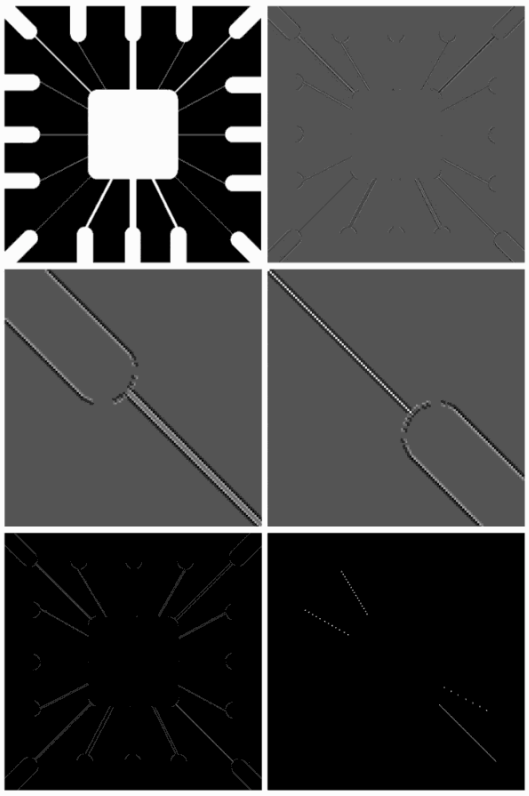
(d) $Q(R_i) = \text{TRUE}$ for $i = 1, 2, \ldots, n$.

(e) $Q(R_i \cup R_j) = \text{FALSE}$ for any adjacent regions R_i and R_j.



a b c
d e f

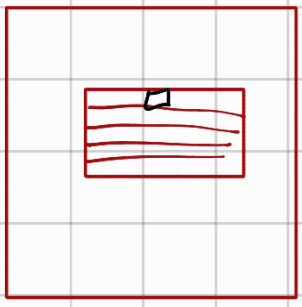
FIGURE 10.1 (a) Image containing a region of constant intensity. (b) Image showing the boundary of the inner region, obtained from intensity discontinuities. (c) Result of segmenting the image into two regions. (d) Image containing a textured region. (e) Result of edge computations. Note the large number of small edges that are connected to the original boundary, making it difficult to find a unique boundary using only edge information. (f) Result of segmentation based on region properties.



a b
c d
e f

FIGURE 10.7
(a) Image of a wire-bond template. (b) Result of processing with the +45° line detector mask in Fig. 10.6. (c) Zoomed view of the top left region of (b). (d) Zoomed view of the bottom right region of (b). (e) The image in (b) with all negative values set to zero. (f) All points (in white) whose values satisfied the condition $g \geq T$, where g is the image in (e). (The points in (f) were enlarged to make them easier to see.)

A third model of an edge is the so-called *roof edge*, having the characteristics illustrated in Fig. 10.8(c). Roof edges are models of lines through a region, with the base (width) of a roof edge being determined by the thickness and sharpness of the line. In the limit, when its base is 1 pixel wide, a roof edge is

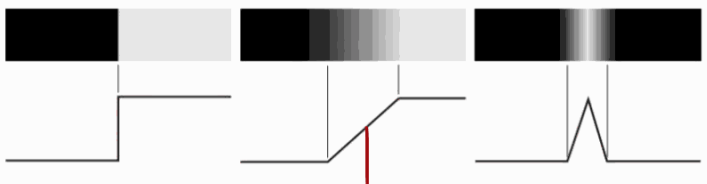


double edge sorunu var

Sınırları belirlemek, bölgeyi bulmak için edge detection segmentation gerekiyor

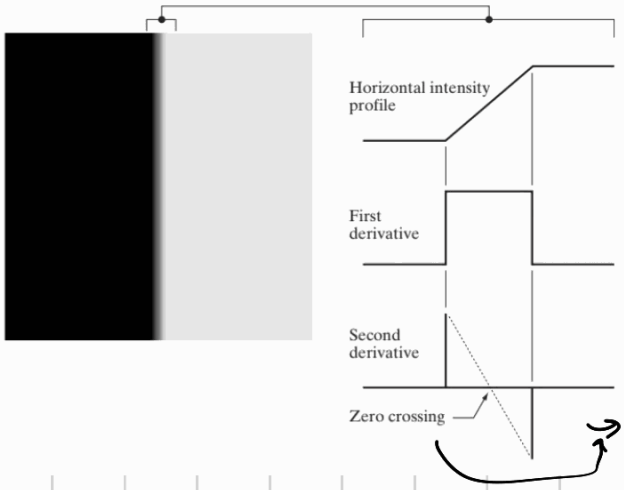
a b c

FIGURE 10.8
From left to right, models (ideal representations) of a step, a ramp, and a roof edge, and their corresponding intensity profiles.



really nothing more than a 1-pixel-thick line running through a region in an

Merkezi bulabiliyor olmamız lazım

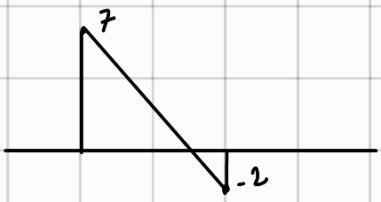


a b

FIGURE 10.10
(a) Two regions of constant intensity separated by an ideal vertical ramp edge.
(b) Detail near the edge, showing a horizontal intensity profile, together with its first and second derivatives.

pozitiften negatife geçişin olduğu yer

7 | -2



birinci dereceden türevinin mutlağı 1'den büyük olmalı

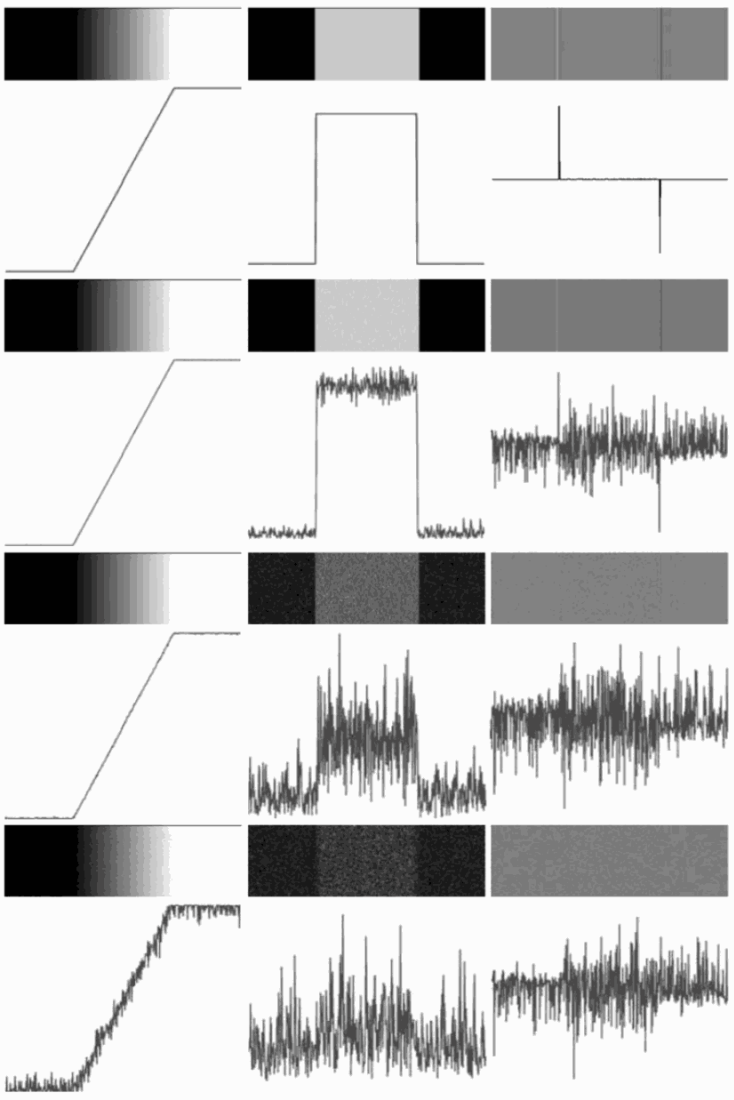


FIGURE 10.11 First column: Images and intensity profiles of a ramp edge corrupted by random Gaussian noise of zero mean and standard deviations of 0.0, 0.1, 1.0, and 10.0 intensity levels, respectively. Second column: First-derivative images and intensity profiles. Third column: Second-derivative images and intensity profiles.

$$\nabla f = \begin{bmatrix} g_x \\ g_y \end{bmatrix} = \begin{bmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{bmatrix}$$

$$M(x,y) = \text{mag}(\nabla f) = (G_x^2 + G_y^2)^{1/2} \approx |G_x| + |G_y|$$

direction of gradient vector

$$\alpha(x,y) = \tan^{-1}\left(\frac{G_y}{G_x}\right)$$

1	2	1
0	0	0
-1	-2	-1

→ x yönü

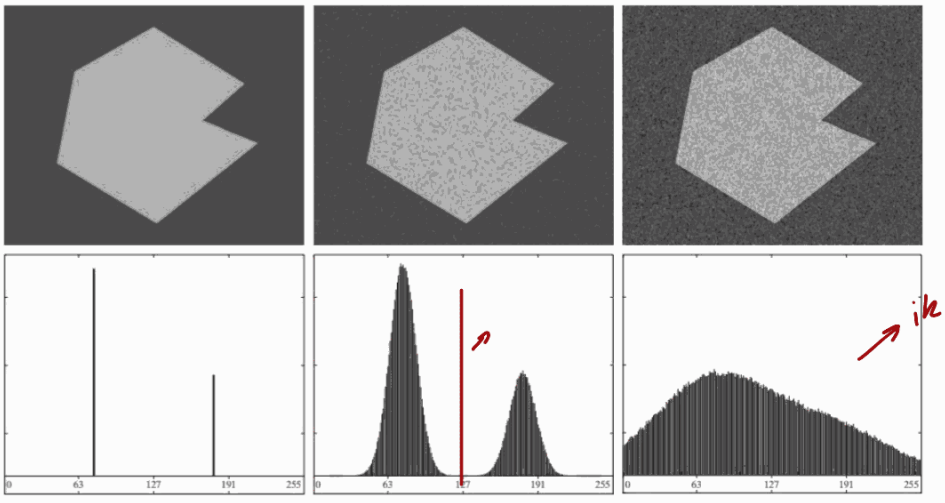
1	0	-1
2	0	-2
1	0	-1

→ y yönü



a b
c d

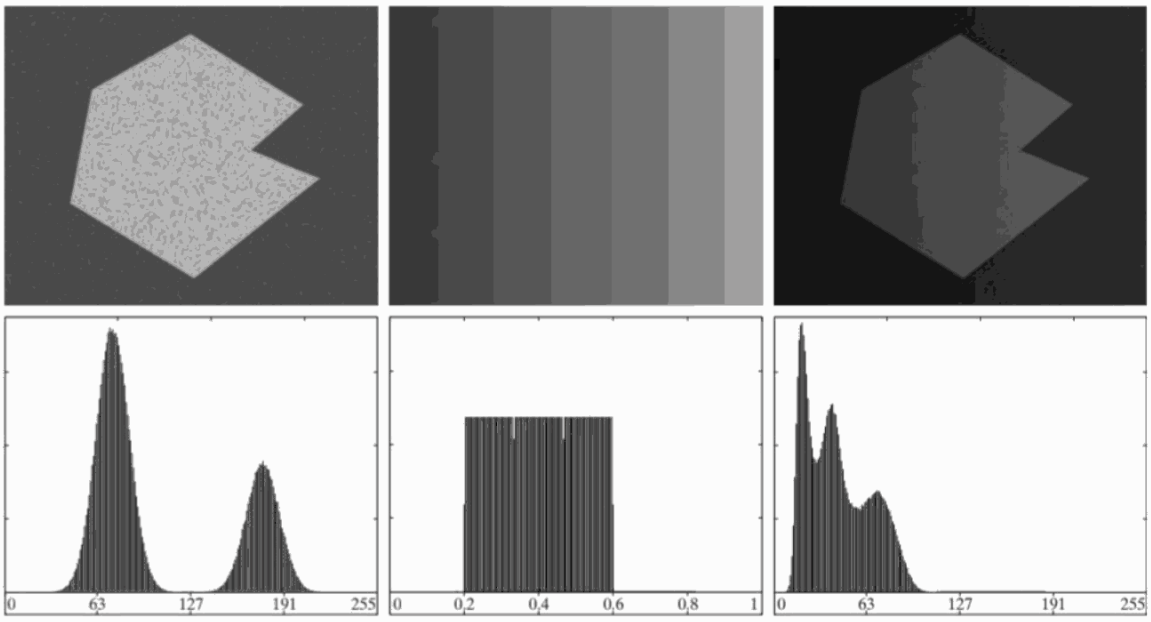
FIGURE 10.16
(a) Original image of size 834 × 1114 pixels, with intensity values scaled to the range [0, 1]. (b) $|g_x|$, the component of the gradient in the x-direction, obtained using the Sobel mask in Fig. 10.14(f) to filter the image. (c) $|g_y|$, obtained using the mask in Fig. 10.14(g). (d) The gradient image, $|g_x| + |g_y|$.



→ ikiagri görüntü var dijenizone

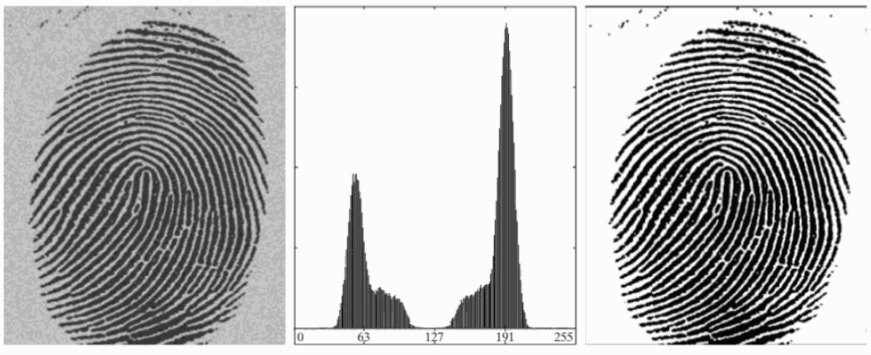
a b c
d e f

FIGURE 10.36 (a) Noiseless 8-bit image. (b) Image with additive Gaussian noise of mean 0 and standard deviation of 10 intensity levels. (c) Image with additive Gaussian noise of mean 0 and standard deviation of 50 intensity levels. (d)–(f) Corresponding histograms.



a b c
d e f

FIGURE 10.37 (a) Noisy image. (b) Intensity ramp in the range [0.2, 0.6]. (c) Product of (a) and (b). (d)–(f) Corresponding histograms.



a b c

FIGURE 10.38 (a) Noisy fingerprint. (b) Histogram. (c) Segmented result using a global threshold (the border was added for clarity). (Original courtesy of the National Institute of Standards and Technology.)

Basic Intensity Tresholding

$$g(x,y) = \begin{cases} 1 & \text{if } f(x,y) > T \\ 0 & \text{O.W} \end{cases}$$

Global tek eşik